

Surge Tanks & Deaerators: Compliance, SOP & Optimization Guide

Purpose:

DA tanks remove oxygen from boiler feedwater, while surge tanks balance supply and protect pumps from low NPSH.

Why It's Ignored:

Internal components are rarely visible, and poor venting or overflow goes unnoticed.

Must-Have Docs:

- Temperature and level log sheets
- Overflow/vent maintenance log
- Steam pressure chart for DA heads
- SOPs for vent, overflow, and draining

Compliance Officials:

ASME Section I, State Boiler Inspector, OSHA 1910.169

PPE Required:

Thermal gloves, faceshield, safety glasses

Installation Practices:

- Steam pressure gauge at tDA head
- Overflow piped to drain
- Vent sized and free-flowing
- Inlet strainers and level control tested

Must Checks:

- DA head steam pressure (5–10 psig typical)
- Tank temperature > 200°F
- Overflow not active during normal ops
- Feedwater pump suction NPSH adequate

Regulations & Codes:

ASME Boiler Code, OSHA, State Mechanical Code